SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct: (192521054)

I K.D. Mirundini *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	1	1 8.5
Concept Mapping Structure	25%	1	1.5 1	1
Linking & Relationships	20%	1.5 1	1	1
Application & Analysis	20%	1	1.5 1	2
Explanation & Clarity	15%	1.5 1	2	1 1.5

Total Marks 24 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress →LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

SIMATS ENGINEERING

ASSESSMENT TOOL - 2

NAME : J.P. Mirunalini Jays

COURSE CODE : CS00710

COURSE NAME : Computer Networks

MISSING LAYERS IN THE OSI MODEL

APPLICATION

Provides network services to end users.

(Eg: Web, email, file Transfer)

PRESENTATION

Handle data formats, encryption compression and Translation

SESSION

Manages sessions keeps connection alive and controls dialogue between endpoint

TRANSPORT

Provides end to end communication, reliability, flow control and error recovery

(Eg: TCP/UDP)

DATA LINK LAYER

Handles Framing Error detection correction MAC addressing and local delivery

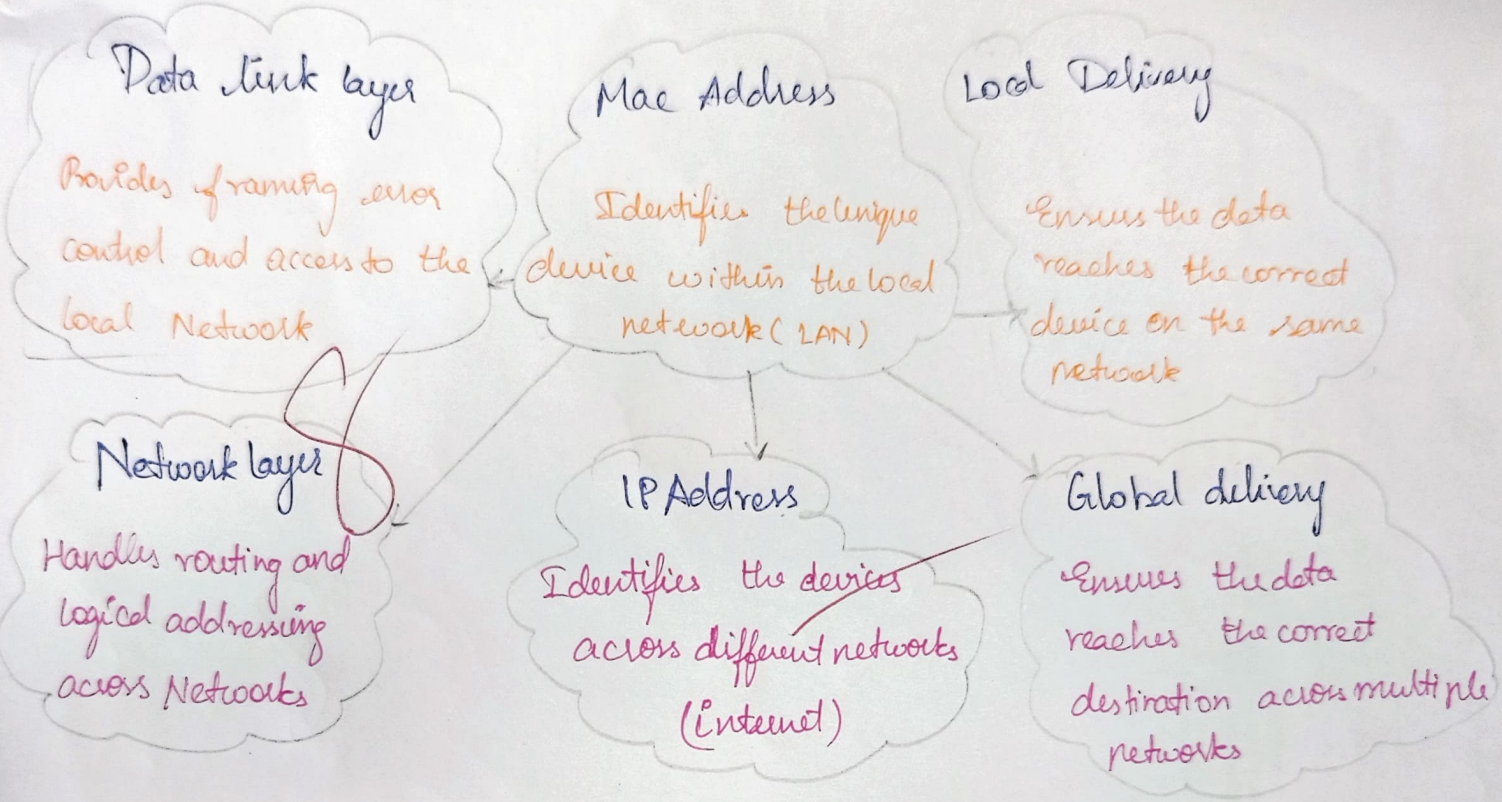
NETWORK

Provides logical addressing and routing of data (eg. IP)

PHYSICAL LAYER

Transmits raw bits as electrical/optical/wireless signals over the medium

Data Link and Network Relationship

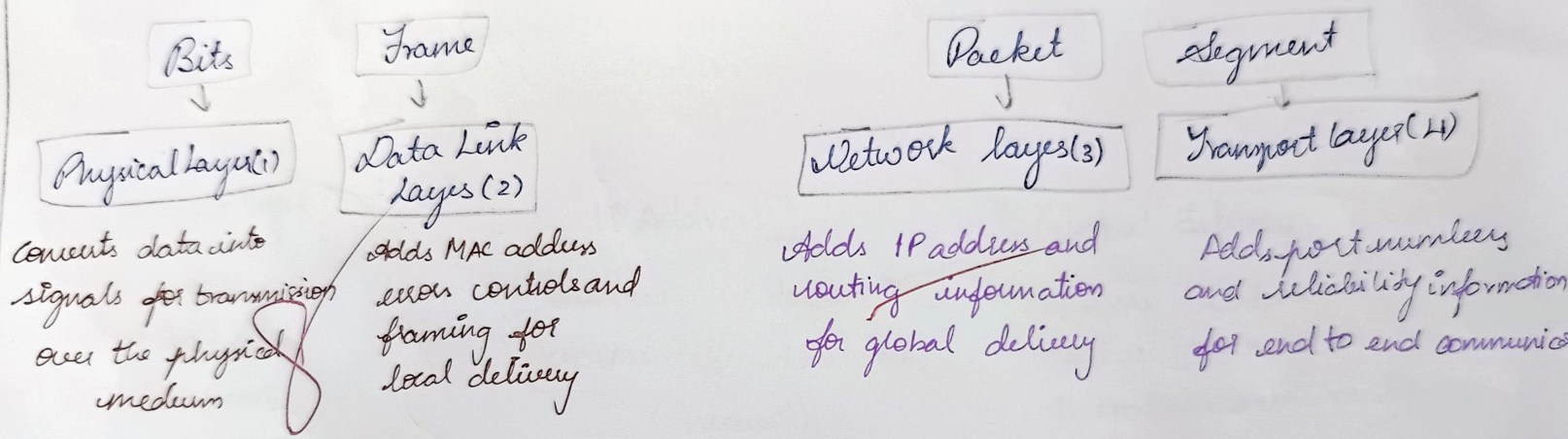


Mac address + Data link layer → ensure the data is delivered to right devices on the local network

IP Address + Network layer → ensure the data is routed across different network to the destination

DATA UNITS AND THEIR CORRESPONDING OSI LAYER

Data Units



How Transformation data units support transmission

1. Segment → Packet : adds network layer information (IP address) for routing
2. Packet → Frame : adds MAC address error control and framing for local delivery

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation